

7.2.2 PSD of Polar Signaling Code

- In polar signaling, 1 is transmitted by a pulse $\mathbf{p(t)}$ and 0 is represented by $-\mathbf{p(t)}$. In this case, a_k is equally likely to be 1 or -1, and a_k^2 is always 1.
- Notice that there are N pulses and $a_k^2 = 1$ for each one, and the summation on the right-hand side of the Eq. of R_0 is equal to N . Hence

$$R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N a_k^2 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N 1 = \lim_{N \rightarrow \infty} \frac{1}{N} N = 1$$

- Moreover, both a_k and a_{k+1} are either 1 or -1. Hence, $a_k a_{k+1}$ is either 1 or -1. Because the pulse amplitude a_k is equally likely to be 1 and -1 on the average, out of N terms the product $a_k a_{k+1}$ is equal to 1 for $N/2$ terms and is equal to -1 for the remaining $N/2$ terms. Therefore

Possible Values of $a_k a_{k+1}$		
$a_k \backslash a_{k+1}$	-1	+1
-1	1	-1
+1	-1	1

$$R_1 = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{2}(1) + \frac{N}{2}(-1) \right] = 0 \quad (7.9)$$

- Arguing this way, we see that the product $a_k a_{k+n}$ is also equally likely to be 1 or -1. Hence,

$$R_n = 0 \text{ for } n \geq 1$$

- Therefore from Eq. (7.7), $S_x(f) = R_o/T_b$

$$\text{and, } S_y(f) = \frac{|P(f)|^2}{T_b} R_o = \frac{|P(f)|^2}{T_b} \quad (7.10)$$

- For RZ code, $p(t)$ is a rectangular pulse of width $T_b/2$ (half-width rectangular pulse), that is,

$$p(t) = \Pi \left(\frac{t}{T_b/2} \right) = \Pi \left(\frac{2t}{T_b} \right)$$

and

$$P(f) = \frac{T_b}{2} \operatorname{sinc} \left(\frac{\pi f T_b}{2} \right) \quad (7.11)$$

By using Eq. (7.10) we therefore get

$$S_y(f) = \frac{T_b}{4} \operatorname{sinc}^2 \left(\frac{\pi f T_b}{2} \right) \quad (7.12)$$

- Note that at $f = 0$, $S_y(f) = \frac{T_b}{4}$
- $S_y(f) = 0$ when $\left(\frac{\pi f T_b}{2} \right) = n\pi$ (A)
- This leads $f_{NULL} = \pm \left(\frac{2n}{T_b} \right) = \pm 2nR_b$ (B)

and $\omega = \pm \left(\frac{4\pi n}{T_b} \right)$ (C)

- Figure 7.6 shows the spectrum $S_y(f)$. It is clear that the polar signal has most of its power concentrated in the lower frequencies.
- Theoretically, the spectrum becomes very small as frequency increases but never becomes totally zero above a certain frequency.
- As given in chapter 3, in order to define a meaningful measure of bandwidth, we consider the first non-dc null frequency of the PSD to be its **essential bandwidth B_s** .
- From Eq. (B), the 1st non-dc null frequency occurs at $n = 1$. Thus, B_s of the RZ polar code is equal to

$$B_s = 2R_b \text{ Hz} \quad (7.13)$$

where R_b is the clock frequency or the pulse rate measured in pulses/ sec.

Recall that the Nyquist bandwidth B_N required to transmit R_b pulses per second is

$$B_N = R_b / 2$$

$$\text{So, } B_s = 2R_b = 4B_N \text{ Hz}$$

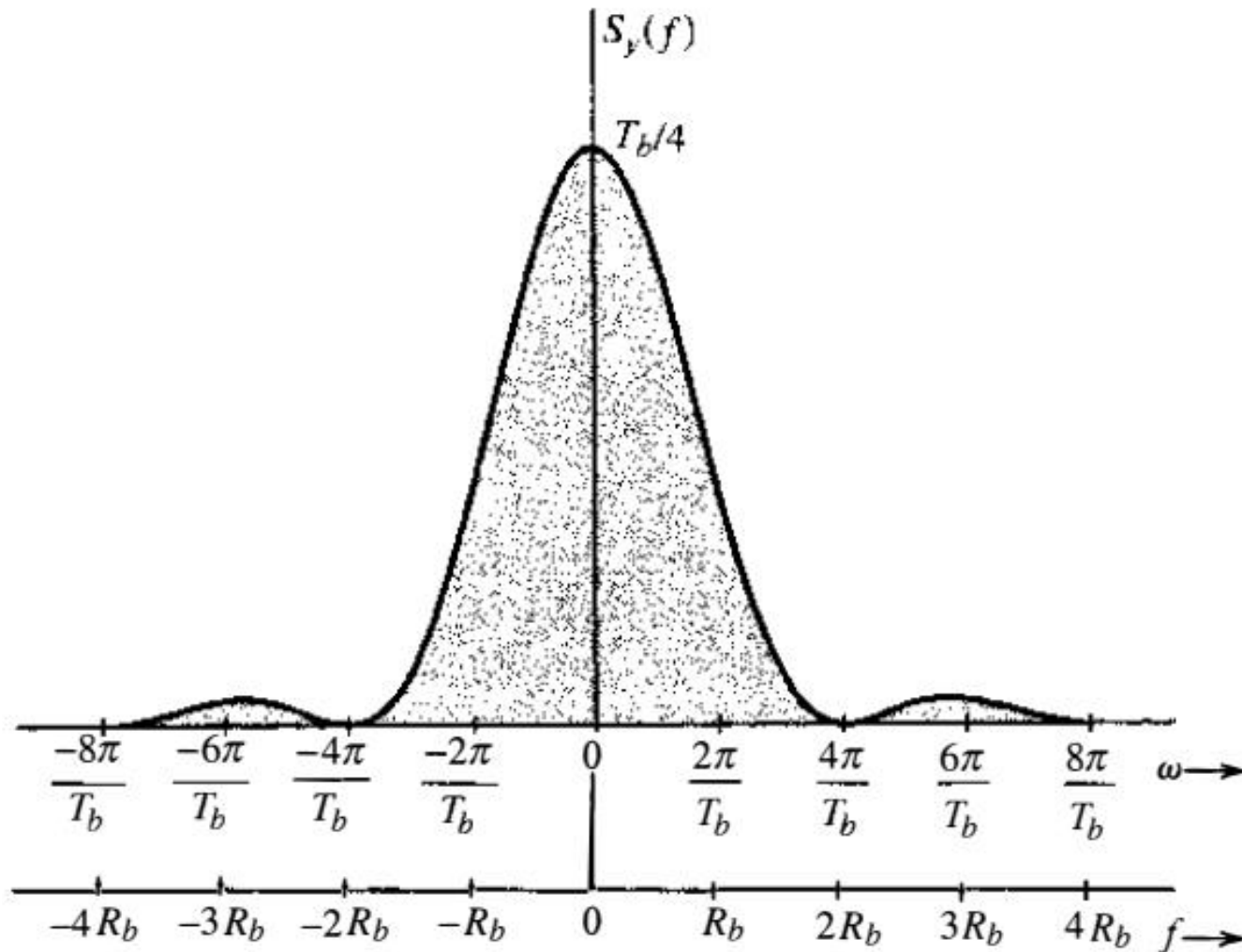


Figure (7.6)
Power
spectral
density (PSD)
of a RZ polar
signal

- Thus for a RZ polar code, B_s is equal to 4 times the theoretical bandwidth (Nyquist bandwidth) required to transmit R_b pulses per second.
 - For a full-width pulse (NRZ pulse), B_s is halved, that is

$$B_s = R_b = 2B_N \text{ Hz} \quad (7.14)$$

- Thus for a NRZ polar code, B_s is equal to 2 times the Nyquist bandwidth.
- So, polar signaling is not the most bandwidth efficient code.
- Second, polar signaling has no capability for self error detection or error correction.
- A third disadvantage of polar signaling is that it has nonzero PSD at dc ($f = 0$).

Advantages of Polar code

- Polar signaling is the most efficient scheme from the power requirement viewpoint. If a pulse $\pm p(t)$ has energy E joule, then because $1/T_b$ digits are transmitted per second, polar signal power is

$$P_p = (E)(1/T_b) = E/T_b \quad \text{Watts} \quad (7.15)$$

- For a given power, it can be shown that the error probability for a polar scheme is the lowest among all signaling techniques.
- Polar signaling is also transparent because there is always some pulse (positive or negative) regardless of the bit sequence. Rectification of the *RZ* polar signal yields a periodic signal of clock frequency and can readily be used to extract timing.

7.2.3 Constructing a DC Null in PSD by Pulse Shaping

- Because the PSD of a line code $[S_y(f)]$ contains a factor $|P(f)|^2$, we can force the PSD to have a dc null by selecting a pulse $p(t)$ such that $P(f)$ is zero at dc ($f = 0$). Since

$$P(f) = \int_{-\infty}^{\infty} p(t)e^{-j2\pi ft} dt \quad , \text{ then } \quad P(0) = \int_{-\infty}^{\infty} p(t) dt$$

- Hence, if the area under $p(t)$ is made zero, $P(0)$ will equal zero, and we have a dc null in the PSD. For a rectangular pulse, one possible shape of $p(t)$ to accomplish this is the **Manchester code**, or **split-phase** (also called **twinned-binary**) shown in Figure (7.7)

- The **split-phase Manchester** code represents **1s** with a positive half interval pulse followed by a negative half-interval pulse, and vice versa for the representation of **0s**. This code is transparent as it guarantees zero DC component regardless of the message sequence. However, it requires an absolute sense of polarity at the receiver.

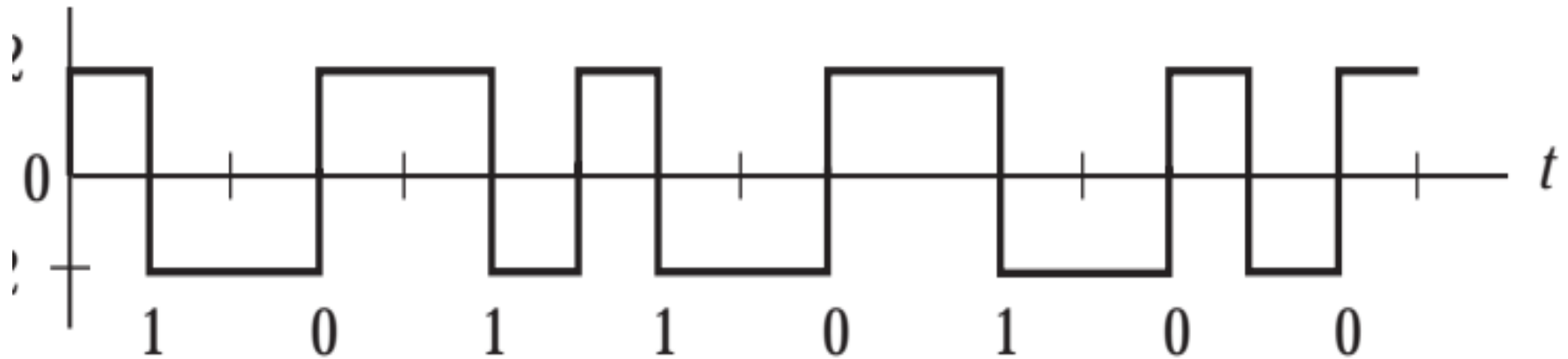


Figure (7.7) The split-phase Manchester

7.2.4 PSD of an On-Off Signaling

- In on-off signaling, 1 is transmitted by a pulse $p(t)$ and 0 is represented by no pulse. Hence, a_k is equally likely to be 1 or 0.
- Out of N pulses in the interval of T seconds, a_k is 1 for $N/2$ pulses and is 0 for the remaining $N/2$ pulses on the average. Hence

$$R_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{2} (1)^2 + \frac{N}{2} (0)^2 \right] = \frac{1}{2} \quad (7.16)$$

- To compute R_n , we need to consider the product $a_k a_{k+n}$. Since a_k and a_{k+n} are equally likely to be 1 or 0, the product $a_k a_{k+n}$ is equally likely to be 1×1 , 1×0 , 0×1 or 0×0 , that is, 1, 0, 0, 0. Therefore on the average, $a_k a_{k+n}$ is equal to 1 for $N/4$ terms and 0 for $3N/4$ terms and

$$R_n = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{4}(1) + \frac{3N}{4}(0) \right] = \frac{1}{4} \quad n \geq 1 \quad (7.17)$$

- Consequently using Eq.(7.6)

$$S_x(f) = \frac{1}{2T_b} + \frac{1}{4T_b} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} e^{-jn2\pi fT_b} \quad (7.18)$$

- With some math, the PSD for the on-off signaling is equal to

$$S_y(f) = \frac{|P(f)|^2}{4T_b} \left[1 + \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T_b}\right) \right] \quad (7.19)$$

- Note that the PSD of the on-off code Eq. (7. 19) has two parts: a continuous part $\left[\frac{|P(f)|^2}{4T_b} \right]$, and a discrete part.

$$\frac{|P(f)|^2}{4T_b^2} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T_b}\right) = \frac{1}{4T_b^2} \sum_{n=-\infty}^{\infty} \left| P\left(\frac{n}{T_b}\right) \right|^2 \delta\left(f - \frac{n}{T_b}\right) \quad (7.20)$$

- This discrete part may be nullified if the pulse shape is chosen such that

$$P\left(\frac{n}{T_b}\right) = 0 \quad n = 0, \pm 1, \dots$$

- For the case of RZ (a half wave rectangular pulse) on-off, the PSD is equal to

$$S_y(f) = \frac{T_b}{16} \text{sinc}^2\left(\frac{\pi f T_b}{2}\right) \left[1 + \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T_b}\right)\right] \quad (7.21)$$

- The continuous component of the spectrum is $(T_b/16) \text{sinc}^2(\pi f T_b/2)$. This is identical (except for a scaling factor of 4) to the spectrum of the polar signal [Eq. (7.12)]. The continuous component of the on-off is exactly half the polar signal discussed earlier. Hence, the **PSD** of this component is one-fourth the **PSD** polar signal.

- The discrete component is represented by the product of an impulse train with the continuous component $(T_b / 16) \text{sinc}^2 (\pi f T_b / 2)$. Hence this component appears as periodic impulses with the continuous component as the envelope. Moreover, the impulses repeat at the clock frequency $R_b = 1/T_b$ because its fundamental frequency is $1/T_b$ Hz. The resulting PSD of the on-off RZ code is shown in Fig. 7.8.

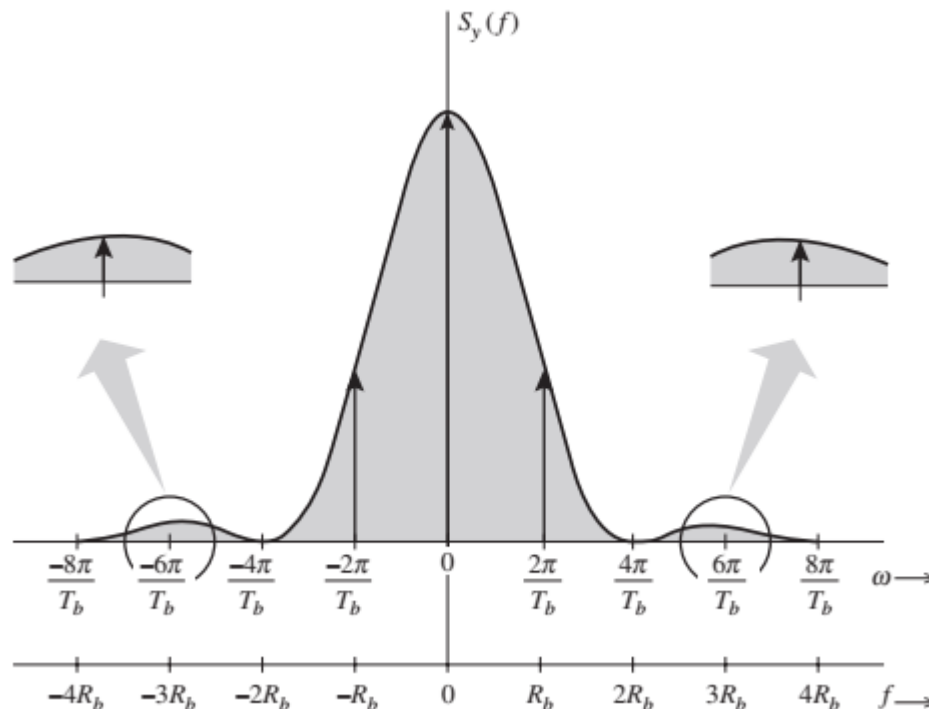


Figure (7.8)
PSD of the
on-off RZ
code

➤ Disadvantage of On-off signaling

- For a given transmitted power, on-off signaling is less immune to noise interference than the polar scheme, which uses a positive pulse for 1 and a negative pulse for 0. This is because the noise immunity depends on the difference of amplitudes representing **1** and **0**. Hence, for the same immunity, if on-off signaling uses pulses of amplitudes **2** and **0**, polar signaling need use only pulses of amplitudes 1 and -1. It is simple to show that on-off signaling requires twice as much power as polar signaling. If a pulse of amplitude 1 or -1 has energy E , then the pulse of amplitude 2 has energy $(2)^2 E = 4E$. So, each pulse energy is $4E$, though on average such a pulse is transmitted over half of the time while nothing is transmitted over the other half.

Hence, the average signal power of on-off is

$$\frac{1}{T_b} \left(4E \frac{1}{2} + 0 \cdot \frac{1}{2} \right) = \frac{2E}{T_b}$$

which is twice that required for the polar signal.

- Moreover, unlike the polar case, on-off signaling is not transparent. A long string of 0s (or offs) causes the absence of a signal and can lead to errors in timing extraction.
- In addition, all the disadvantages of polar signaling, (e.g., excessive transmission bandwidth, nonzero power spectrum at dc, no error detection (or correction) capability are also present in on-off signaling.

7.2.5 PSD of Bipolar (AMI or pseudoternary) Signaling

- Bipolar code is used in PCM for telephone networks.
- A 0 is transmitted by no pulse, and a 1 is transmitted by a pulse $p(t)$ or $-p(t)$, depending on whether the previous 1 was transmitted by $-p(t)$ or $p(t)$. With consecutive pulses alternating, we can avoid dc wander and thus cause a dc null in the PSD.
- Bipolar signaling actually uses three symbols [$p(t)$, 0, and $-p(t)$], and hence, it is in reality ternary rather than binary signaling.
- To calculate the PSD, we have
$$R_o = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_k a_k^2$$
- On the average, half of the a_k s are 0, and the other half are either 1 or -1, with $a_k^2 = 1$. Therefore,

$$R_o = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{2} (\pm 1)^2 + \frac{N}{2} (0)^2 \right] = \frac{1}{2}$$

- To compute R_1 , we consider the pulse strength product $a_k a_{k+1}$. There are four equally likely sequences of two bits: 11, 10, 01, 00. Since bit 0 is encoded by no pulse ($a_k = 0$), the product $a_k a_{k+1}$ is zero for the last three of these sequences. This means, on the average, that $3N/4$ combinations have $a_k a_{k+1} = 0$ and only $N/4$ combinations have nonzero $a_k a_{k+1}$. Because of the bipolar rule, the bit sequence 11 can be encoded only by two consecutive pulses of opposite polarities. This means the product $a_k a_{k+1} = -1$ for the $N/4$ combinations. Therefore

Possible Values of $a_k a_{k+1}$

$a_k \backslash a_{k+1}$	0	1
0	0	0
1	0	-1

$$R_1 = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{4} (-1) + \frac{3N}{4} (0) \right] = -\frac{1}{4}$$

- To compute R_2 in a similar way, we need to observe the product $a_k a_{k+2}$. For this, we need to consider all possible combinations of three bits in sequence. There are eight equally likely combinations: 111, 101, 110, 100, 011, 010, 001, 000. The last six combinations have either the first and/or the last bit 0. Hence $a_k a_{k+2} = 0$ for all these six combinations. The first two combinations are the only ones that yield nonzero $a_k a_{k+2}$. From the bipolar rule, the first and the third pulses in the combination 111 are of the same polarity, yielding $a_k a_{k+2} = 1$. But for 101, the first and the third pulse are of opposite polarity, yielding $a_k a_{k+2} = -1$. Thus, on the average, $a_k a_{k+2} = 1$ for $N/8$ terms, and $= -1$ for $N/8$ terms, and 0 for $3N/4$ terms. Hence,

$$R_2 = \lim_{N \rightarrow \infty} \frac{1}{N} \left[\frac{N}{8}(1) + \frac{N}{8}(-1) + \frac{3N}{8}(0) \right] = 0$$

- For $n > 2$, the product $a_k a_{k+n}$ can be 1, -1, or 0. Moreover, an equal number of combinations have values 1 and -1.

Thus $R_n = 0$ for $n > 1$.

and [see Eq.(7.8)]

$$S_y(f) = \frac{|P(f)|^2}{T_b} [1 - \cos 2\pi f T_b] = \frac{|P(f)|^2}{T_b} \sin^2(\pi f T_b) \quad (7.22)$$

- Note that $S_y(f) = 0$ for $f = 0$ (dc), regardless of $P(f)$. Hence, the PSD has a dc null, which is desirable for ac coupling. Moreover, $\sin^2(\pi f T_b) = 0$ at $f = 1/T_b$, that is, at $f = 1/T_b = R_b$ Hz. Thus, regardless of $P(f)$, we are assured of the first non-dc null bandwidth is R_b Hz as shown in Fig. (7.9)

➤ For the half-width pulse (RZ)

$$S_y(f) = \frac{T_b}{4} \text{sinc}^2\left(\frac{\pi f T_b}{2}\right) \sin^2(\pi f T_b)$$

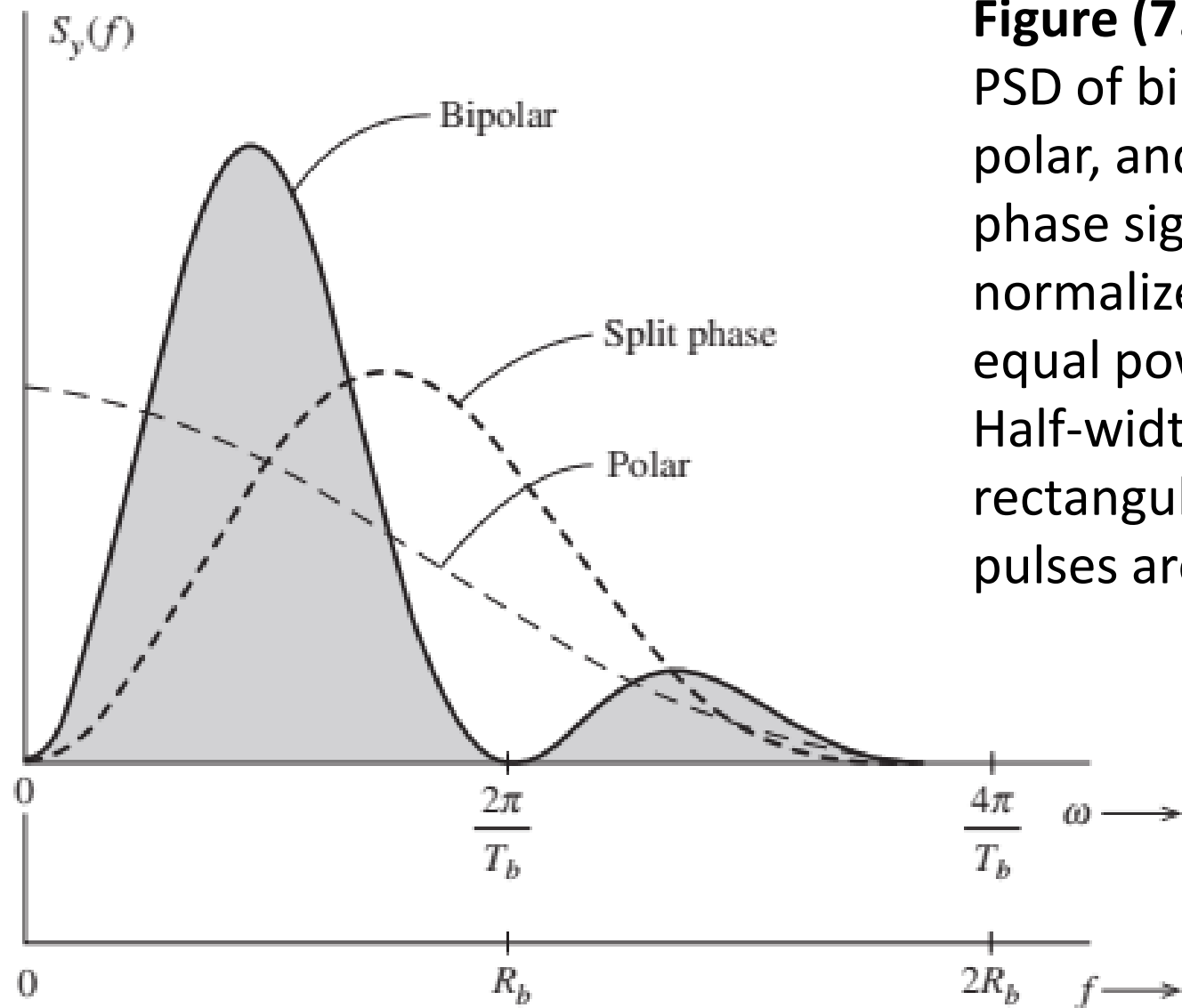


Figure (7.9)
PSD of bipolar, polar, and split-phase signals normalized for equal powers. Half-width rectangular pulses are used

- The essential bandwidth of the signal is R_b ($R_b = 1/T_b$), which is half that of polar using the same half-width pulse or on-off signaling and twice the theoretical minimum bandwidth. Observe that we were able to obtain the bandwidth R_b for polar (or on-off) case for full-width pulse. For the bipolar case, the bandwidth is R_b Hz whether the pulse is half-width or full-width.
- Bipolar signaling has several advantages:
 - (1) its spectrum has a non dc null; (2) its bandwidth is not excessive; (3) it has single-error-detection capability. This is because a single error will cause a violation of the alternating pulse rule, and this will be immediately detected.
 - (4) If a bipolar signal is rectified, we get an on-off signal that has a discrete component at the clock frequency.